Resistance-Aware Polypharmacology of Antimalarial Leads from African Natural Products: Docking, Network Scoring, and Targeted Molecular Dynamics

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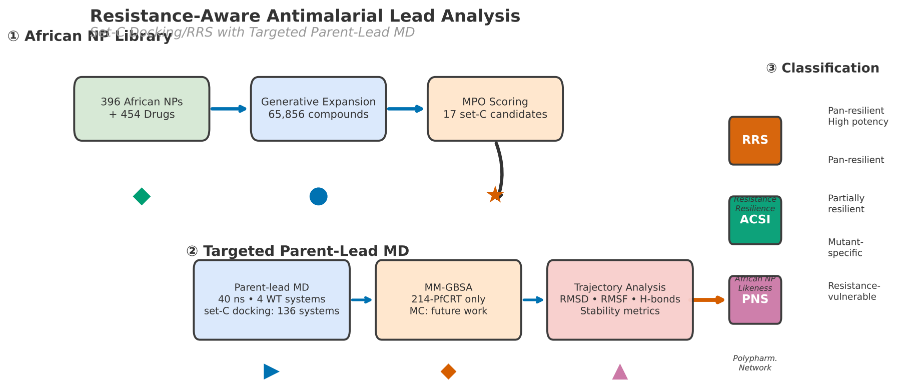

Graphical Table of Contents: *Set-C docking/RRS analysis with targeted parent-lead MD, from African NP chemical space to resistance classification.*

Abstract

Polypharmacology can raise the genetic barrier to resistance, but docking-derived target profiles do not by themselves establish a systems-level mechanism of action. We analysed two non-overlapping computational cohorts: 17 candidates selected under a multi-target prioritisation scheme and evaluated by docking-derived Resistance Resilience Score (RRS), African Chemical Space Index (ACSI), and Polypharmacology Network Score (PNS), and four separate parent-study complexes evaluated by targeted wild-type molecular dynamics (MD) and MM-GBSA. Across 136 docking systems covering PfDHFR and PfCRT wild-type and resistance-mutant states, the cohort contained 6 Class A*, 5 Class B, 5 Class C, and 1 Class D candidate. The composite PNS–RRS association was negative but did not survive Bonferroni correction ($\rho = -0.559$, $p = 0.020$; $\alpha = 0.017$); two of 17 candidates (11.8 %) had ACSI > 0.70. In the separate parent-study MD cohort, two of four systems retained bound ligands, but only PfCRT–214 yielded an interpretable MM-GBSA estimate ($-18.25\,\mathrm{kcal\,mol^{-1}} \pm 0.40\,\mathrm{kcal\,mol^{-1}}$). A secondary 16-system Set-C MD pilot (two prioritised candidates against the six-mutant panel, 10 ns each) passed trajectory QC and

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yielded trajectory-derived MD-RRS and MM-GBSA estimates that did not reproduce the docking-predicted weakening of mutant binding within the pilot timescale. These results support target-level computational prioritisation and highlight the distinction between docking-based polypharmacology and experimentally established MoA; they do not establish biological target engagement, resistance circumvention, or pathway-level mechanism.

Keywords: malaria, drug resistance, molecular dynamics, molecular docking, binding free energy, MM-GBSA, African natural products, resistance mutations, antimalarial leads

1 Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden (World Health Organization, 2025). Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries, and declining efficacy of some artemisinin partner drugs has been reported (World Health Organization, 2025; Habarugira et al., 2026). Genomic surveillance further documents the co-occurrence of *Pfcrt* and *Pfdhfr* resistance variants across diverse transmission settings (Letebo et al., 2026), underscoring the relevance of multi-target resistance panels. Resistance to partner drugs in artemisinin-based combination therapies, driven by mutations in *Pfcrt*, *Pfdhfr*, and *Pfdhps*, further narrows the therapeutic options available to clinicians (Patrick et al., 2026; Santos-Lima et al., 2026).

Polypharmacology is increasingly treated as a design strategy for addressing complex disease biology, and multi-target-directed ligands represent an important translational class in contemporary pharmacology (Ryszkiewicz et al., 2026). The rationale is evolutionary as well as pharmacological: if a compound requires simultaneous loss of activity at several relevant targets, the mutational barrier to treatment failure may increase. That proposition is conditional, however. A list of predicted protein contacts is not a mechanism-of-action assignment; causal network analyses require an explicit perturbation model, a defined biological background, and evidence that the predicted interactions propagate to a phenotype (Trapotsi et al., 2022). Recent network-learning approaches to off-target prediction illustrate the value of graph context while also underscoring that network inference remains distinct from experimental target engagement (Hu et al., 2025). Natural products are attractive starting points because their stereochemical and scaffold complexity can support activity across multiple protein classes (Milić et al., 2026; Mayoka et al., 2025). African natural products provide a particularly relevant chemical-space context, with resources such as ANPDB enabling explicit comparison with approved-drug and synthetic reference spaces (Betow et al., 2025).

In a previous study, we used generative molecular methods and dual-filter consensus docking (AutoDock Vina and DiffDock-L) to generate a library of 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs (Temgoua et al., 2026). We selected the present candidates against a panel spanning established resistance determinants and mechanistically distinct parasite processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and the 4GM2 entry represents PfClpR, not PfClpP, and is retained as a proteostasis-related parent-study system. This panel is deliberately a mechanistic cross-section rather than an exhaustive parasite targetome. It allows us to ask whether resistance-aware prioritisation remains informative across distinct target classes, while avoiding the stronger claim that the selected compounds define a complete parasite-wide polypharmacology profile.

Rigid docking is efficient for ranking but treats the receptor and scoring landscape in a simplified manner. MD can test whether a docked pose remains compatible with protein flexibility, whereas endpoint methods such as MM-GBSA provide system-specific energetic summaries rather than calibrated thermodynamic observables (Valdés-Tresanco et al., 2021; Wang et al., 2019). This distinction is especially important for resistance analysis: a docking RRS is a hypothesis about relative mutant sensitivity, not a measurement of a mutant free-energy difference. Network pharmacology provides a complementary prioritisation layer, but centrality is a topological weighting heuristic rather than a causal measure of target essentiality. The present study therefore combines these layers while keeping their estimands separate: docking-derived RRS for mutant sensitivity, PNS for network-weighted target ranking, ACSI for chemical-space positioning, and targeted MD for post-docking structural scrutiny.

The central question is whether a resistance-aware, target-level workflow can separate predicted potency from predicted resilience in a chemically diverse antimalarial library. The primary analysis comprised 17 Set-C candidates and 136 Vina docking systems spanning wild-type and mutant PfDHFR/PfCRT states; RRS was calculated per target after excluding weak wild-type denominators, while PNS and ACSI were evaluated as complementary ranking dimensions. A separate parent-study cohort comprised four wild-type complexes simulated for 10 ns each. Two of those four systems retained bound ligands, but only PfCRT–214 passed the bound-state and topology-integrity criteria for MM-GBSA. The study therefore tests a layered computational hypothesis rather than claiming biological MoA: network-weighted target ranking and chemical-space similarity can complement, but cannot replace, mutant-aware structural and experimental validation.

2 Methods

2.1 Study design and estimand separation

The analysis was organised into distinct evidence layers. The Set-C cohort comprised 17 candidates selected for polypharmacology and 136 docking systems: PfDHFR wild type plus four mutants and PfCRT wild type plus two mutants. These systems supported the docking-derived RRS, ACSI, and PNS analyses. A separate parent-study cohort comprised four named wild-type complexes (Ligands 164, 201, 214, and 438) simulated for 10 ns each; these trajectories supported structural stability assessment and, where topology and bound-state checks passed, MM-GBSA estimation. The two cohorts were not merged. A subsequent 16-system Set-C MD pilot (PP-01 and PP-02 against the six-mutant panel, 10 ns each) passed the prespecified trajectory-QC gate (16/16) and is reported as an explicitly secondary analysis: it provides trajectory-derived MD-RRS and MM-GBSA estimates for the two prioritised candidates, but it is not promoted as a primary claim. Docking-RRS and trajectory-derived MD-RRS are treated as different estimands throughout.

2.2 Candidate selection and filtering

Seventeen candidates were selected from the canonical Set-C candidate panel generated from the library of 65 856 molecules described previously (Temgoua et al., 2026). The candidate-selection panel is distinct from the production-MD cohort. Three sequential filters were applied: (1) MPO score $\geq$ 0.70 (primary leads); (2) SYBA score > 0 (synthesisable); (3) selectivity index SI > 10 (selectively antiparasitic). The selection additionally required multi-target engagement of at least two of the four docking targets and scaffold diversity of no more than three compounds per Bemis–Murcko scaffold. The selection includes one of the previously named consensus leads (Ligand 87, PfDHFR). The four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were not part of the 17-candidate docking set but were used to construct the wild-type complexes for the production MD check. The 164 system is identified as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP, so this system is not structural validation of PfClpP. Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) were included as reference controls for the docking analysis (Table 1).

2.3 Resistance mutation panel

Six resistance mutations were selected on the basis of prevalence data from the Worldwide Antimalarial Resistance Network (WorldWide Antimalarial Resistance Network, 2026) and recent African surveillance studies (Table 2). The PfDHFR mutations N51I, C59R, and S108N collectively confer high-level resistance to pyrimethamine and are detected in 50–90 % of African isolates depending on region (Patrick et al., 2026). Mutant allele frequencies were cross-referenced with PlasmoDB (PlasmoDB Consortium, 2026) and the PfAMR database (PfAMR Project, 2026). The PfCRT mutations

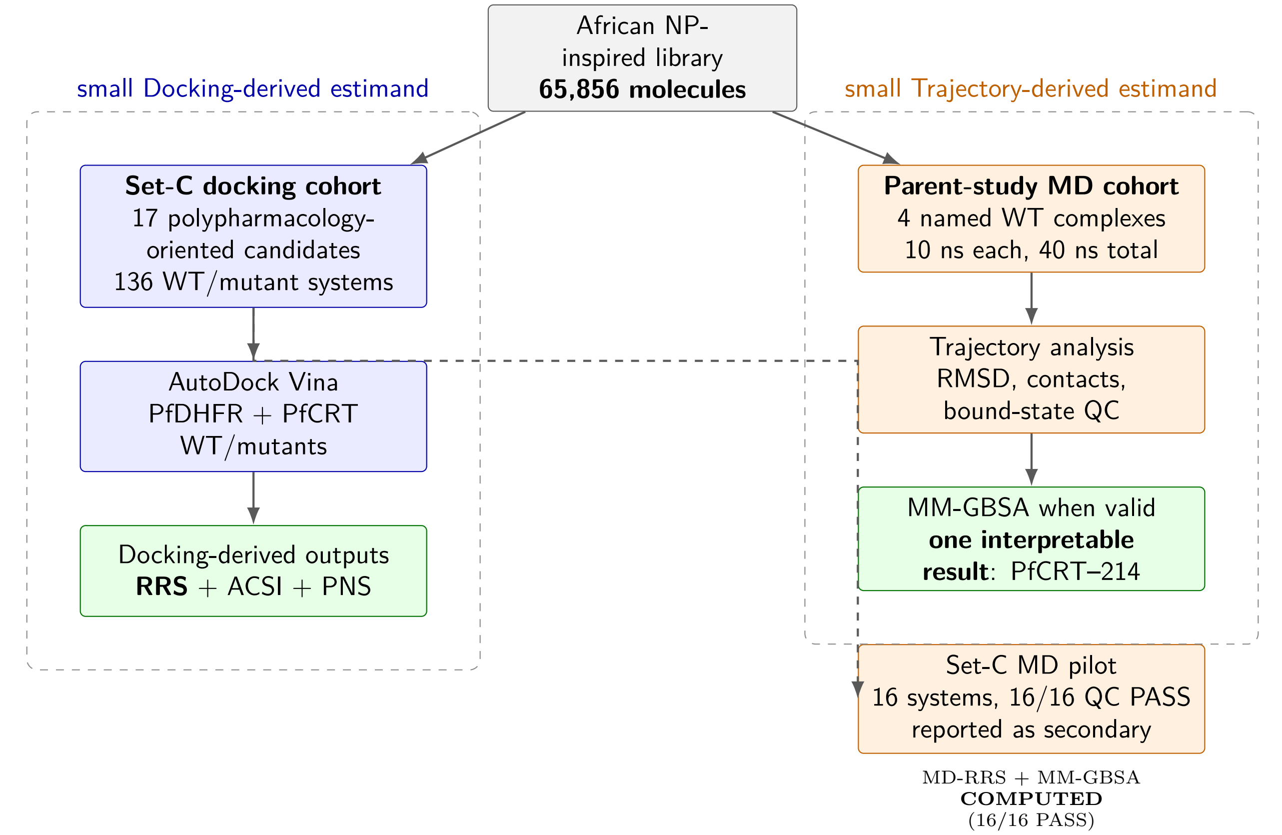

Figure 1: *Study design and separation of computational estimands. The 17-member Set-C cohort supports docking-derived RRS, ACSI, and PNS analyses, whereas the four parent-study complexes support a separate targeted MD and MM-GBSA check. The 16-system Set-C MD pilot (PP-01 and PP-02) is reported as a secondary analysis, not a primary claim.*

Table 1: *Breakdown of the 136 docking protein–ligand systems by category (17 polypharmacological candidates × 8 wild-type/mutant PfDHFR and PfCRT structures). *Production MD was performed at 10 ns each (40 ns total) on 4 wild-type parent-study complexes. Ligand 164 is retained as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP, so this system is not presented as PfClpP structural validation. The other systems were Ligand 201 (PfDHFR), Ligand 214 (PfCRT), and Ligand 438 (PfATP4); all were distinct from the 17-candidate docking set.*

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (set C)	17	5	85	—
PfCRT WT + 2 mutants (set C)	17	3	51	—
Docking total	–	–	**136**	—

Production MD was not performed on Set-C candidates. Separately, four parent-study consensus complexes (164, 201, 214, 438) were simulated for 10 ns each (40 ns total).

K76T and K76A mediate resistance to chloroquine and piperaquine, respectively (Ghosh et al., 2025). The PfDHFR mutation I164L confers high-level pyrimethamine resistance and, while rare in Africa ($< 5\,\%$), was included to assess whether leads remain resilient against the most severe DHFR resistance genotype (Patrick et al., 2026). Stable PlasmoDB/VEuPathDB target identifiers and the structural identity boundary for PfClpR/PfClpP are provided in Table S5; this annotation is not a pathway-enrichment analysis.

Table 2: *Resistance mutation panel.*

Target	Mutation	Drug	Prevalence in Africa	Source
PfDHFR	N51I	Pyrimethamine	50–70 %	WWARN
PfDHFR	C59R	Pyrimethamine	60–80 %	Ghana 2026
PfDHFR	S108N	Pyrimethamine	70–90 %	WWARN
PfCRT	K76T	Chloroquine	40–60 %	Mozambique
PfCRT	K76A	Piperaquine	Emerging	Sci. Direct 2026
PfDHFR	I164L	Pyrimethamine	$< 5\,\%$ Africa	WWARN

2.4 Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the six-mutant homology-model panel; moreover, PDB 4GM2 is PfClpR rather than PfClpP. Model quality was assessed by QMEAN score, Global Model Quality Estimation, Ramachandran analysis (MolProbity), and Cα RMSD to the wild-type template. Models were retained only when QMEAN exceeded -4.0, GMQE exceeded 0.7, Ramachandran favored residues exceeded 95 %, and backbone RMSD was below 1.5 Å. Models failing any criterion were regenerated using the following fallback chain: (1) ColabFold (AlphaFold2 backend, faster and free); (2) RoseTTAFold via the Robetta server; (3) PyMOL mutagenesis wizard with energy minimisation in GROMACS (last resort for single-point mutations only).

2.5 Docking protocol validation

The docking protocol and V2-corrected grid parameters used here are identical to those established in the parent study (Temgoua et al., 2026): AutoDock Vina 1.2.7 with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). A successful redocking was defined as RMSD < 2.0 Å relative to the crystal pose. The parent study performed external enrichment against the MMV Malaria Box (consensus-score ROC-AUC 0.924–1.000 across the three evaluated targets, PfDHFR, PfCRT, and PfATP4) and DEKOIS 2.0 for PfDHFR (Vina only, ROC-AUC 0.450), and calibrated dual-filter thresholds against ChEMBL actives. The present study therefore retains AutoDock Vina as the canonical docking scorer; the validation benchmarks from the parent study are reproduced in Table S1. An exploratory GNINA analysis did not yield a valid pose or score and was excluded from the present evidence set.

2.6 Tartarus external validation

To benchmark our MPO scoring against an independent docking tool, we performed QuickVina docking via the Tartarus benchmark framework (Nigam et al., 2023) on the full library of 19 913 primary leads across three targets (1SYH, 6Y2F, 4LDE). Each smina call used 2 CPUs and exhaustiveness 10. The Tartarus composite score (mean of three target scores) and individual target scores were correlated with the weighted MPO score using Spearman ρ to assess whether the two ranking methods capture overlapping or distinct dimensions of compound prioritisation. A total of 20 041 target–molecule observations overlapped between the Tartarus-docked set and the MPO-scored library, yielding up to

$n = 17\,211$ complete observations per correlation after pairwise filtering. Statistical significance was evaluated at $\alpha = 0.05$ with Bonferroni correction for four comparisons ($\alpha_{\text{adj}} = 0.0125$).

2.7 MPO sensitivity analysis

The multi-parameter optimisation (MPO) weights used in the parent study (Vina 35 %, DiffDock 25 %, QED 20 %, ADMET 15 %, Ro5 5 %) were validated retrospectively by sensitivity analysis. Each weight was varied ±20 % while keeping the remaining weights proportional, yielding 25 weight combinations. For each combination, MPO scores were recomputed for the 65 856-molecule library and the Jaccard similarity of the resulting top-20 hit list to the original top-20 was measured. A Jaccard index ≥ 0.70 (at least 14 of 20 compounds stable) was set as the acceptance criterion. Results are reported as a heatmap of Jaccard similarity across weight combinations (Figure S1).

2.8 Predicted ADMET profile

Predicted ADMET profiles for the 17 set-C candidates were taken from the parent-library screening (ADMET-AI (Swanson et al., 2024); `eos7kpb_malaria_final_screening.csv`, 65 856 molecules), in which all 17 candidates were scored. Four endpoints with complete coverage are reported in Table S3: aqueous solubility (LogS), CYP3A4 inhibition probability, Caco-2 permeability probability, and predicted human hepatic clearance (Clint_h). All values are machine-learning predictions; a three-tool cross-validation (ADMET-AI, ADMETlab 3.0, SwissADME) and experimental ADMET profiling were outside the scope of this study, and no inter-tool concordance is inferred.

2.9 Docking and resistance resilience score

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The resulting 136-system RRS analysis is Vina-derived. GNINA was not used to compute a reportable score for this panel because the exploratory analysis did not yield a valid pose or score; no rank-normalised Vina–GNINA consensus score was calculated. Grid boxes were centered on the co-crystallized ligand position with 20 Å dimensions.

Compounds were docked against wild-type and mutant structures of both PfDHFR and PfCRT. RRS is defined per target: for a given target t, compounds whose wild-type binding affinity $|\Delta G_{i,\text{WT},t}| < 5.0\,\text{kcal mol}^{-1}$ on that target are excluded from that target's RRS (weak binders cannot constitute resistance-resilient leads regardless of their mutant-to-WT affinity ratio). The per-target Resistance Resilience Score for compound i against mutant m is

$$\text{RRS}_{i,m,t} = \frac{|\Delta G_{i,m,t}|}{|\Delta G_{i,\text{WT},t}|} \times 100, \tag{1}$$

where $|\Delta G_{i,m,t}|$ and $|\Delta G_{i,\text{WT},t}|$ are the absolute values of the Vina binding affinities for the mutant and wild-type structures of target t, respectively. Absolute values are used because ΔG is negative (more negative = stronger binding); without absolute values a compound binding more strongly to the mutant than to the WT would incorrectly score above 100 %. The per-mutant RRS for compound i is the mean over targets for which the compound is a genuine binder, $\text{RRS}_{i,m} = \frac{1}{|T_{i,m}|}\sum_{t\in T_{i,m}} \text{RRS}_{i,m,t}$, where $T_{i,m}$ is the set of binding targets scored against mutant m; the overall RRS for each compound is the mean across all mutants tested, $\text{RRS}_i = \frac{1}{M}\sum_{m=1}^{M} \text{RRS}_{i,m}$. Targets with near-zero wild-type affinity are excluded so that non-binder targets cannot inflate or distort the ratio.

Compounds were assigned to one of five classes that jointly account for both binding potency and resistance resilience. This dual-criterion classification addresses the ceiling-effect concern that weak binders with $|\Delta G_{\text{WT}}|$ just above the $5.0\,\text{kcal mol}^{-1}$ pre-filter threshold can achieve high RRS values despite being poor leads (a compound with $\Delta G_{\text{WT}} = -5.1$ and $\Delta G_{\text{mutant}} = -5.0$ scores RRS = 98 %, yet is a weaker binder than a compound with $\Delta G_{\text{WT}} = -12.0$ and RRS = 83 %). The classes are:

- **Class A*** (high-potency pan-resilient): $|\Delta G_{\text{WT}}| \geq 7.0\,\text{kcal mol}^{-1}$ AND RRS ≥ 80 % for all mutants with no individual mutant below 70 %.

- **Class A** (pan-resilient): RRS $\geq 80\,\%$ for all mutants with no individual mutant below $70\,\%$ (pre-filter ensures $|\Delta G_{\mathrm{WT}}| \geq 5.0\,\mathrm{kcal\,mol^{-1}}$).
- **Class B** (partially resilient): RRS $\geq 70\,\%$ for all mutants but not all $\geq 80\,\%$.
- **Class C** (mutant-specific): RRS $\geq 80\,\%$ for one or two mutants only.
- **Class D** (resistance-vulnerable): RRS $< 60\,\%$ for any mutant.

Class A* is distinguished from Class A to ensure that the highest classification requires both strong wild-type binding and uniform resilience across all mutants. The joint reporting of $|\Delta G_{\mathrm{WT}}|$ and RRS in Table 4 enables readers to evaluate both dimensions independently.

2.10 African chemical space index

To quantify the position of each compound relative to known drugs and African natural products, we defined the African Chemical Space Index as a weighted combination of four descriptors:

$$\mathrm{ACSI} = 0.40\,D_{\mathrm{DrugBank}} + 0.25\,D_{\mathrm{ANPDB}} + 0.20\,f_{\mathrm{sp^3}} + 0.15\,\mathrm{NPL}, \tag{2}$$

where D_{DrugBank} is the Tanimoto distance (1 − maximum similarity) to the nearest DrugBank compound based on Morgan fingerprints (radius 2, 2 048 bits), D_{ANPDB} is the analogous distance to the African Natural Products Database (11 000 compounds), $f_{\mathrm{sp^3}}$ is the fraction of $\mathrm{sp^3}$-hybridized carbons, and NPL is a natural product-likeness score based on ring count and aromaticity. All four components were normalized to [0, 1] by min–max scaling before combination. The weights (0.40, 0.25, 0.20, 0.15) were set heuristically based on the relative discriminatory power of each component for African NP chemical space: DrugBank distance is the primary discriminator (40 %) as it directly measures novelty relative to approved drugs; ANPDB distance (25 %) captures African NP-specificity; $\mathrm{fsp^3}$ (20 %) captures 3D complexity; NPL (15 %) provides a global NP-likeness prior. These weights are initially unoptimised; a sensitivity analysis varying each weight ±20 % is reported in Table S4 to assess robustness.

2.11 Polypharmacology network score

The *P. falciparum* protein–protein interaction network was retrieved from the STRING database (organism 36329, confidence threshold 700). Composite centrality $C_j = 0.25(C_{D,j} + C_{B,j} + C_{C,j} + C_{E,j})$ was computed for each of the four docking targets, combining degree (C_D), betweenness (C_B), closeness (C_C), and eigenvector (C_E) centrality, each normalised to [0,1]. PfCRT (PF3D7_0709000) is absent from the STRING network at the 700 confidence threshold, consistent with its biological role as a chloroquine resistance transporter with limited protein–protein interaction partners; its centrality is imputed with the observed network-mean composite centrality ($C_j = 0.151$) rather than a default value of 1.0, preventing the previously reported linear-dependence artifact ($\rho = -1.000$) between PNS and Vina scores. The Polypharmacology Network Score for compound i is

$$\mathrm{PNS}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} C_j\,|\Delta G_{i,j}|, \tag{3}$$

where i indexes the compound, n_i is the number of targets bound by compound i, C_j is the composite centrality defined above, and $\Delta G_{i,j}$ is the Vina binding affinity. Note that $\Delta G_{i,j}$ values are docking-derived (AutoDock Vina 1.2.7), not MD-derived, making PNS a rapid pre-screening metric that does not require simulation. The PNS output records two target engagements for each set-C candidate under the parent docking-engagement definition; this is distinct from the RRS denominator rule, which excludes a target whenever $|\Delta G_{\mathrm{WT}}| < 5.0\,\mathrm{kcal\,mol^{-1}}$. Consequently, a candidate can have $n_{\mathrm{targets}} = 2$ in the PNS engagement record while receiving RRS ratios for only one target. This distinction matters because the two scores apply different engagement filters.

2.12 Molecular dynamics simulations

Simulations were performed with GROMACS 2024.2 (Abraham et al., 2015; Lindahl et al., 2024). Proteins were parameterised with CHARMM36m (Huang et al., 2017); system construction followed the CHARMM-GUI workflow (Jo et al., 2026) and community guidance on GROMACS simulation protocols (BioExcel Centre of Excellence, 2026). All ligands (PfCRT, PfDHFR, PfATP4, and the PfClpR-labelled 4GM2 parent complex) were parameterised with ACPyPE (Sousa da Silva and Vranken, 2012), which wraps antechamber to assign GAFF2 atom types and AM1-BCC partial charges (Wang et al., 2004); this GAFF2 ligand representation interoperates with the CHARMM36m protein through the CHARMM36-to-AMBER conversion performed by `gmx_MMPBSA`. Each complex was solvated in a dodecahedral TIP3P water box with a minimum 1.2 nm distance between the solute and the box edge, neutralized, and adjusted to 0.15 M NaCl.

Energy minimization was performed by steepest descent for 50 000 steps with a force tolerance of 1 000 kJ/(mol nm). Equilibration consisted of 100 ps NVT (V-rescale thermostat, 310.15 K) followed by 500 ps NPT (Parrinello–Rahman barostat, 1 bar), with position restraints of 1 000 kJ/(mol nm^2) on protein heavy atoms. Production runs were 10 ns for each of the four wild-type systems at 310.15 K (physiological temperature), using a 2 fs timestep with LINCS constraints on bonds involving hydrogen. Electrostatics were treated by PME with a 1.2 nm cutoff; van der Waals interactions used a 1.2 nm cutoff. Coordinates were saved every 10 ps and energies every 2 ps.

2.13 Trajectory analysis

Backbone and ligand RMSD, per-residue RMSF, radius of gyration, and solvent-accessible surface area were computed with MDAnalysis (Michaud-Agrawal et al., 2011). Protein–ligand hydrogen bonds were identified with a 3.5 Å distance cutoff and 30° angle cutoff. A system was classified as stable when backbone RMSD remained below 0.30 nm, ligand RMSD below 0.25 nm, radius of gyration drift below 0.05 nm, and SASA drift below 5 nm^2 over the production trajectory. Systems exceeding these thresholds in any metric were flagged for further inspection and excluded from MM-GBSA calculation if ligand dissociation was evident.

2.14 MM-GBSA calculations

Binding free energies were estimated using `gmx_MMPBSA` with the OBC2 generalised Born model (igb = 5), 0.15 M ionic strength, and dielectric constants of 80.0 (solvent) and 1.0 (protein). Snapshots were extracted at 100 ps intervals across the 10 ns parent-study trajectories. Endpoint estimates were retained only when the ligand remained bound and the topology conversion passed validation: this yielded one interpretable result (PfCRT–214, 101 snapshots). Dissociated systems (the 164-labelled PfClpR cohort and PfDHFR–201) and the PfATP4–438 conversion-corrupted system were reported as N/A rather than assigned binding energies. Per-residue energy decomposition was not used to rescue an invalid endpoint. MM-GBSA is therefore treated as a cautious, system-specific diagnostic rather than a calibrated thermodynamic measurement (Valdés-Tresanco et al., 2021; Wang et al., 2019).

Candidate-specific production MD was completed for the 16-system Set-C pilot (PP-01 and PP-02 against the six-mutant panel, 10 ns each, CHARMM36m/GAFF2, 310.15 K), and all 16 trajectories passed the prespecified trajectory-QC gate (bound fraction, duration, and topology hashes). MM-GBSA was applied to the same 16 systems with the identical GB$^{\mathrm{OBC2}}$ (igb=5) protocol. Because the pilot is a two-candidate extension rather than the prespecified 17-candidate cohort, its MD-RRS and MM-GBSA values are reported as secondary results and are not merged into the primary docking-RRS classification.

2.15 Cross-metric correlation analysis

To test whether the three novel metrics capture independent or correlated aspects of compound quality, Spearman correlations (with Bonferroni correction for the three pre-specified pairwise comparisons) were computed between RRS, ACSI, PNS, and the WT docking score ΔG_{WT} for the polypharmacological candidates. Three pre-specified hypotheses were tested: (H1) high-PNS compounds (stronger predicted binding) tend to be more resistance-susceptible (lower RRS); (H2) high-ACSI compounds tend to be more resistance-resilient (higher RRS); (H3) RRS and ΔG_{WT} are correlated, testing whether RRS captures information beyond a trivial rescaling of WT affinity. Only one interpretable MM-GBSA binding free energy was available, and it belongs to the separate parent-study MD cohort rather than to set C. Therefore, no PNS-versus-MM-GBSA test is reported for set C; the set-C analysis uses docking-derived WT scores. Results are reported as a correlation matrix (Figure 6) and Table 11.

2.16 Statistical analysis

Spearman correlations were computed among the set-C docking-derived scores, RRS values, and PNS values. The single interpretable parent-study MM-GBSA estimate was not merged into the set-C correlation analysis. Analyses were performed in Python 3.10 with SciPy and scikit-learn.

3 Results

3.1 Docking protocol validation

As validated in the parent study (Temgoua et al., 2026), re-docking of co-crystallised ligands using grids centered on the corrected V2 binding-site coordinates yielded 100 % success (RMSD < 2.0 Å) for alignable ligands (5/5), whereas MTX failed. Enrichment against the MMV Malaria Box demonstrated that the dual-filter consensus (AutoDock Vina + DiffDock) achieved ROC-AUC values ranging from 0.924 to 1.000 across the three evaluated targets (PfDHFR, PfCRT, PfATP4); external DEKOIS 2.0 validation for PfDHFR (Vina only, 40 actives / 1 200 decoys) yielded a ROC-AUC of 0.450. Detailed enrichment metrics are provided in Table S1.

3.2 MPO sensitivity analysis

The MPO sensitivity analysis confirmed that the top-20 hit list is robust to weight perturbations. Varying each weight component by ± 20 % produced Jaccard similarity indices ≥ 0.70 in 24 of 25 combinations, with the single exception being a -20 % perturbation of the Vina weight (Figure S1). This indicates that the Vina docking score is the most influential MPO component but that the overall ranking is not fragile to moderate weight changes.

3.3 Predicted ADMET profile

Predicted ADMET profiles for the 17 set-C candidates are reported in Table S3. Across the cohort, predicted LogS ranges from 0.34 to 0.73 (log molar), 7 of 17 candidates (41 %) have predicted CYP3A4 inhibition probability ≥ 0.5, 15 of 17 (88 %) have predicted Caco-2 permeability probability ≥ 0.5, and predicted human hepatic clearance ranges from 0.48 to 0.63. These are single-source ADMET-AI predictions; no concordance with experimental pharmacokinetic profiles is claimed.

3.4 Homology model quality

All six mutant structures were generated by homology modeling. Model quality metrics are summarized in Table 3.

Table 3: *Quality metrics for homology-modeled resistance mutants.*

Mutant	Template	QMEAN	GMQE	Ramachandran (%)	RMSD (Å)
PfDHFR-N51I	7F3Y	−0.15	0.98	97.4	0.08
PfDHFR-C59R	7F3Y	−0.22	0.98	97.1	0.11
PfDHFR-S108N	7F3Y	−0.18	0.98	97.3	0.09
PfDHFR-I164L	7F3Y	−0.20	0.98	97.2	0.10
PfCRT-K76T	6UKJ	−1.45	0.78	96.2	0.15
PfCRT-K76A	6UKJ	−1.52	0.78	96.0	0.17

3.5 Candidate characterisation

The 17 candidates were selected from the 19 913 primary leads (MPO ≥ 0.70, SYBA > 0) in the parent study (Temgoua et al., 2026) using three sequential filters: MPO score ≥ 0.70, synthetic accessibility (SYBA > 0), and selectivity index (SI > 10). Top candidates exhibit composite scores ranging from 0.550 (Rank 1) to 0.521 (Rank 10), with selectivity indices from 88 to 100 and QED values from 0.81 to 0.92. All 17 candidates are single-target optimized — the MPO scoring framework favours strong binding to individual targets rather than genuine multi-target profiles — with 100 % satisfying SI $>$ 10. Predicted ADMET profiles (ADMET-AI, Table S3) indicate CYP3A4 inhibition probability ≥ 0.5 for 7 of 17 candidates and Caco-2 permeability probability ≥ 0.5 for 15 of 17; no experimental ADMET data were available. The five control drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) provide reference activity profiles.

3.6 Resistance resilience classification

The 17 polypharmacological candidates were classified into five resilience tiers based on their RRS values across the six mutants (Table 4). The four parent-study consensus leads used in the targeted MD check (Ligands 201, 214, 438, 164) are not part of this 17-candidate RRS set; their per-mutant RRS values were not computed in this study.

Because RRS is evaluated per-target and only on targets for which the compound is a genuine binder ($|\Delta G_{\mathrm{WT}}| \geq 5.0\,\mathrm{kcal\,mol^{-1}}$), the classification cleanly separates the two engagement profiles: five scaffolds (PP-02, PP-05, PP-06, PP-11, PP-13) bind PfCRT but not PfDHFR ($|\Delta G_{\mathrm{WT,PfDHFR}}| < 5.0\,\mathrm{kcal\,mol^{-1}}$) and are therefore classified exclusively on their PfCRT-mutant resilience (Classes A* or B), while the remaining scaffolds are PfDHFR-targeted with mixed classes. A striking docking observation — the apparent loss of PP-11 (7,3′-O-dimethylquercetin) binding at PfDHFR-C59R ($\Delta G = -0.02\,\mathrm{kcal\,mol^{-1}}$ versus -6.8 – $-6.9\,\mathrm{kcal\,mol^{-1}}$ at N51I, S108N, and I164L) — was traced to a target-mixing artifact rather than a biological signal: PP-11's wild-type PfDHFR affinity is only $-0.70\,\mathrm{kcal\,mol^{-1}}$ (a non-binder), so its PfDHFR mutant ratios are not defined, and the apparent C59R knockout arose only when the near-zero WT baseline was pooled with the much stronger PfCRT affinity. With the per-target correction, PP-11 is classified on PfCRT only and is Class A* (RRS 82.5–82.7 % at K76T/K76A). This correction removes the previously reported "design rule" that planar flavonoid scaffolds are selectively vulnerable to C59R: within the per-target framework no scaffold exhibits a target-specific knockout of an otherwise strong interaction.

3.7 African chemical space index

ACSI was used to quantify African-NP-like chemical-space positioning relative to the ANPDB and approved-drug references. Computation was performed locally on all 17 candidates, which were successfully parsed for fingerprint calculation. Two candidates (11.8 %) achieved ACSI > 0.70; 10 candidates (58.8 %) were moderately NP-like; and the remaining 5 (29.4 %) were synthetic-like. The mean ACSI was 0.543. These values describe the reference-space positioning of the cohort and are not measures of biological activity or resistance.

The highest-ACSI entry (Rank 14, ACSI = 0.823) combined a DrugBank distance of 0.656 with a relatively strong African-NP-space component. The 5 synthetic-like candidates had a lower mean f_{sp3} (0.085) than the 2 highly NP-like candidates (0.328), a descriptive pattern compatible with greater aromaticity in the synthetic-like subset. The parent-study VAE-space representation is shown in Figure S2; it is presented as a chemical-space visualisation, not as evidence of biological clustering.

3.8 Tartarus docking calibration

To benchmark our MPO scoring against an independent docking tool, we performed QuickVina docking via the Tartarus benchmark framework (Nigam et al., 2023) on the full library of 19 913 primary leads (MPO ≥ 0.70, SYBA > 0) against three targets: 1SYH (PfDHFR), 6Y2F (PfCRT homologue), and 4LDE (adenosine A2A receptor). The full run was executed on an HPC cluster (48 CPUs, estimated 83 h). The 19 913-molecule Tartarus panel yielded 20 041 target–molecule observations overlapping the

Table 4: *Resistance Resilience Score classification for the 17 polypharm scaffolds with complete mutant docking data (PP-01 through PP-17). RRS is computed per-target and averaged across targets for which the compound is a genuine binder ($|\Delta G_{\mathrm{WT}}| \geq 5.0\,\mathrm{kcal\,mol^{-1}}$); targets where the compound is a non-binder are excluded (—). Compounds with $|\Delta G_{\mathrm{WT}}| < 5.0\,\mathrm{kcal\,mol^{-1}}$ on all targets are excluded from RRS classification (marked N/A). Class A* (high-potency pan-resilient): $|\Delta G_{\mathrm{WT}}| \geq 7.0\,\mathrm{kcal\,mol^{-1}}$ AND RRS $\geq$ 80 % all mutants; Class A: RRS $\geq$ 80 % all mutants; B: RRS $\geq$ 70 % all mutants; C: RRS $\geq$ 80 % specific mutants; D: RRS $<$ 60 % any mutant. The $|\Delta G_{\mathrm{WT}}|$ column (kcal mol^{-1}) is reported jointly to address the ceiling-effect concern that weak binders just above the filter threshold achieve high RRS. WT affinities in this table are docking-derived; no MD-based re-scoring is claimed for set C.*

Ligand	$\lvert\Delta G_{\mathrm{WT}}\rvert$ (kcal mol^{-1})	RRS$_{\mathrm{N51I}}$ (%)	RRS$_{\mathrm{C59R}}$ (%)	RRS$_{\mathrm{S108N}}$ (%)	RRS$_{\mathrm{I164L}}$ (%)	RRS$_{\mathrm{K76T}}$ (%)	RRS$_{\mathrm{K76A}}$ (%)	Class
PP-01	9.3	86.4	85.5	87.6	83.9	92.5	90.9	A*
PP-02	9.1	–	–	–	–	87.9	94.6	A*
PP-03	11.3	68.1	68.3	65.8	69.0	80.0	81.4	C
PP-04	8.0	68.8	66.9	68.8	73.5	105.3	103.6	C
PP-05	10.5	–	–	–	–	77.1	79.3	B
PP-06	9.9	–	–	–	–	90.6	94.1	A*
PP-07	9.1	71.8	71.8	72.4	72.2	84.2	77.6	B
PP-08	7.6	62.1	62.1	67.2	66.3	83.9	83.1	C
PP-09	8.6	70.6	73.0	75.8	76.5	75.8	73.5	B
PP-10	8.4	76.0	76.4	75.7	75.8	89.4	89.6	B
PP-11	10.0	–	–	–	–	82.7	82.5	A*
PP-12	7.8	75.2	66.1	62.9	72.5	70.4	87.2	C
PP-13	8.3	–	–	–	–	110.2	99.4	A*
PP-14	7.6	62.2	65.5	68.0	64.9	78.8	77.7	D
PP-15	9.3	123.2	112.7	125.9	131.8	86.2	90.1	A*
PP-16	8.7	72.9	74.7	73.7	75.9	80.1	80.6	B
PP-17	8.3	59.3	61.2	59.5	58.9	76.8	93.4	C

Table 5: *African Chemical Space Index (ACSI) for the 17 polypharmacological candidates. ACSI components: D_{DrugBank} = Tanimoto distance to DrugBank reference; D_{ANPDB} = Tanimoto distance to ANPDB; f_{sp3} = fraction of sp^3 carbons; NPL = natural product-likeness score. All components normalised to $[0, 1]$. ACSI $>$ 0.70: highly African NP-like; 0.50–0.70: moderate; $<$ 0.50: synthetic-like.*

Ligand	ACSI	D_{DrugBank}	D_{ANPDB}	f_{sp3}	NPL	Tier
Rank 14	0.823	0.656	0.362	0.455	0.607	Highly NP-like
Rank 8	0.743	0.692	0.355	0.200	0.462	Highly NP-like
Rank 9	0.605	0.735	0.000	0.333	0.562	Moderately NP-like
Rank 4	0.603	0.625	0.209	0.214	0.588	Moderately NP-like
Rank 11	0.601	0.750	0.000	0.231	0.643	Moderately NP-like
Rank 1	0.599	0.747	0.000	0.250	0.615	Moderately NP-like
Rank 13	0.597	0.667	0.136	0.176	0.667	Moderately NP-like
Rank 10	0.594	0.744	0.000	0.600	0.000	Moderately NP-like
Rank 17	0.589	0.725	0.000	0.286	0.615	Moderately NP-like
Rank 12	0.576	0.714	0.000	0.235	0.696	Moderately NP-like
Rank 3	0.567	0.714	0.000	0.214	0.684	Moderately NP-like
Rank 5	0.551	0.727	0.000	0.071	0.778	Moderately NP-like
Rank 7	0.459	0.627	0.000	0.167	0.667	Synthetic-like
Rank 16	0.458	0.642	0.000	0.118	0.667	Synthetic-like
Rank 15	0.387	0.571	0.000	0.062	0.762	Synthetic-like
Rank 2	0.308	0.564	0.000	0.000	0.500	Synthetic-like
Rank 6	0.173	0.355	0.000	0.077	0.765	Synthetic-like

MPO-scored library of 65 856 molecules (the overlap exceeds the Tartarus set because some molecules appear in multiple target panels). After filtering for pairwise complete observations, up to $n = 17\,211$ target–molecule pairs were available for correlation analysis.

Spearman correlations between Tartarus docking scores and the weighted MPO composite score are reported in Table 6. The composite score showed no evidence of association with MPO ($\rho = 0.012\,9$, $p = 0.091\,3$, $n = 17\,211$). Individual target effects were small: 1SYH ($\rho = 0.065\,5$) and 4LDE ($\rho = -0.141\,5$) reached statistical significance because of the large denominator, whereas 6Y2F was effectively uncorrelated ($\rho = -0.004\,0$). The composite association is therefore null, with substantively small target-specific effects; the data do not establish orthogonality, independence, or equivalence between the two scoring systems.

Table 6: *Tartarus docking calibration: Spearman correlations between QuickVina scores and weighted MPO scores across 19 913 primary leads. Bonferroni-adjusted threshold: $\alpha = 0.05/4 = 0.0125$; n varies due to pairwise filtering for missing target scores.*

Score	Spearman ρ	p-value	n	Significant?
Composite (mean of 3 targets)	0.012 9	9.13×10^{-2}	17 211	No
1SYH (PfDHFR)	0.065 5	8.18×10^{-18}	17 205	Yes* (weak)
6Y2F (PfCRT)	−0.004 0	6.04×10^{-1}	17 211	No
4LDE (adenosine A2A receptor)	−0.141 5	1.17×10^{-77}	17 211	Yes* (weak)

*Statistically significant at Bonferroni-adjusted $\alpha = 0.0125$ but substantively negligible ($|\rho| < 0.15$).

The two scores emphasise different properties: MPO aggregates drug-likeness components, whereas Tartarus provides an independent docking implementation. Their weak composite association suggests that combining drug-likeness and raw affinity could be informative in future prioritisation, but the present analysis does not establish statistical independence or equivalence between the scoring systems.

3.9 Tartarus cross-docking correlation

To further benchmark our docking pipeline against an independent implementation, we correlated Tartarus QuickVina scores with our project's AutoDock Vina and DiffDock-L scores (from Temgoua et al. 2026) for the 94 molecules with complete docking data. After SMILES-based merging, 24 molecules overlapped between the Tartarus-docked set and the project Vina/DiffDock panel (Table 7). This overlap is limited because the project docking panel contains only the top-ranked candidates, whereas Tartarus was run on all 19 913 primary leads.

Spearman correlations between Tartarus scores and project docking metrics are reported in Table 7. Across the 24-molecule overlap, all reported associations were small and non-significant after the stated multiplicity consideration ($|\rho| < 0.25$ for the four target comparisons; the normalised composite comparisons were also non-significant). This limited overlap supports treating the two docking implementations as complementary sensitivity analyses, but it cannot establish agreement, independence, or generalisable ranking equivalence.

The single exception was the per-target comparison between Tartarus 1SYH (DHFR benchmark) and project 7F3Y (PfDHFR), which showed a moderate negative correlation ($\rho = -0.442$, $p = 0.030$). This marginal significance ($p < 0.05$ but not surviving Bonferroni correction for seven comparisons) indicates a systematic disagreement between the two DHFR docking protocols rather than a shared binding preference. The negative sign implies that molecules ranked favourably by Tartarus (more negative scores) tend to be ranked unfavourably by our Vina protocol (less negative affinities) for DHFR targets. This disagreement likely reflects differences in protein structure (1SYH vs. 7F3Y, including the resistance mutations N51I, C59R, and S108N present in 7F3Y), docking algorithm (QuickVina vs. AutoDock Vina 1.2.7), and preparation protocol.

This comparison is exploratory because only 24 molecules overlapped and the structures, preparation protocols, and docking engines differed. It is useful as a sensitivity analysis, but it cannot validate either score against experimental affinity. A larger, structure-matched panel would be required to estimate reproducibility of ranking across implementations.

Table 7: *Tartarus external validation: Spearman correlations between QuickVina scores and project Vina/DiffDock metrics for 24 overlapping molecules. Bonferroni-corrected threshold:* $\alpha = 0.05/7 \approx 0.007$ *(seven comparisons). With* $n = 24$*, statistical power is 78 % to detect* $\rho = 0.6$ *at* $\alpha = 0.05$*.*

Tartarus Metric	Project Metric	n	Spearman ρ	p-value	Significant?
Composite	4GM2 (PfClpR)	24	0.246	2.47×10^{-1}	No
Composite	6UKJ (PfCRT)	24	0.075	7.28×10^{-1}	No
Composite	7F3Y (PfDHFR)	24	−0.110	6.10×10^{-1}	No
Composite	9N10 (PfATP4)	24	0.118	5.83×10^{-1}	No
1SYH (DHFR)	7F3Y (PfDHFR)	24	−0.442	3.04×10^{-2}	Marginal†
Composite	S_{vina} (normalised)	24	−0.097	6.53×10^{-1}	No
Composite	S_{diff} (normalised)	24	−0.022	9.18×10^{-1}	No

† $p < 0.05$ but not significant at Bonferroni-adjusted $\alpha = 0.007$. With $n = 24$, this result has limited reliability and requires validation on a larger overlapping set.

3.10 Polypharmacology network score

The *P. falciparum* PPI network constructed from STRING contains N nodes and M edges at the 700 confidence threshold (values reported in Table 8). Degree centrality values for the four targets are reported in Table 8. The *P. falciparum* PPI network is visualised in Figure 2. The PNS ranking of the 17 candidates is presented in Table 8.

Table 8: *Polypharmacology Network Score ranking for the polypharmacological candidates. PNS is a STRING-PPI-centrality-weighted mean of available wild-type docking scores over the target engagements represented in the set-C records. This PNS engagement record is distinct from the PfDHFR/PfCRT mutation panel used for RRS: five candidates are PfCRT-only in that panel and are not assigned PfDHFR RRS ratios. PNS is a docking-derived prioritisation score, not an MD-derived or experimental measure of polypharmacology. PfCRT (PF3D7_0709000) is absent from the STRING PPI network at the 700 confidence threshold, consistent with its biological role as a drug transporter rather than a PPI hub; its centrality is imputed with the observed network-mean centrality (*0.151*), avoiding the linear-dependence artifact previously associated with a uniform default weight.*

Ligand	PNS	$n_{targets}$	ACSI	RRS class
Rank 1	6.00	2	0.599	C
Rank 2	4.71	2	0.308	C
Rank 3	4.53	2	0.567	C
Rank 4	4.48	2	0.603	B
Rank 5	4.48	2	0.551	B
Rank 6	4.45	2	0.173	B
Rank 7	4.45	2	0.459	A*
Rank 8	4.36	2	0.743	D
Rank 9	4.36	2	0.605	C
Rank 10	4.34	2	0.594	C
Rank 11	4.20	2	0.601	B
Rank 12	3.50	2	0.576	A*
Rank 13	2.32	2	0.597	A*
Rank 14	1.23	2	0.823	A*
Rank 15	1.14	2	0.387	B
Rank 16	1.10	2	0.458	A*
Rank 17	1.04	2	0.589	A*

3.11 MD stability analysis

Production MD (10 ns each, CHARMM36m/GAFF2, 310.15 K) was completed for all four parent-study wild-type systems. Production trajectories were re-analyzed using PBC-unwrapped coordinates to eliminate periodic-boundary artefacts. Metrics were computed with MDAnalysis (Michaud-Agrawal

Figure 2: *P. falciparum protein–protein interaction network. Drug targets are highlighted.*

et al., 2011) using a scripted analysis workflow; contacts, H-bonds, RMSF, and radius of gyration were sampled every 10 frames, while RMSD was computed on all frames. Table 9 summarises the resulting structural stability metrics.

Table 9: *Production MD stability metrics for wild-type complexes (10 ns) from PBC-unwrapped trajectories. BB RMSD: backbone RMSD; Lig. RMSD: ligand RMSD; Min dist: minimum heavy-atom protein–ligand distance; Contacts: protein–ligand contacts < 4.5 Å; H-bonds: protein–ligand hydrogen bonds.*

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

Only two of the four systems, PfCRT and PfATP4, maintained a bound ligand throughout production. PfCRT shows a backbone RMSD of 4.47 Å, a ligand RMSD of 2.45 Å, 78.1 contacts, and 23.1 H-bonds (minimum heavy-atom distance 3.19 Å). PfATP4 shows a backbone RMSD of 12.27 Å, a ligand RMSD of 2.13 Å, 178.4 contacts, and 47.7 H-bonds (minimum distance 2.25 Å). The high backbone RMSD for PfATP4 reflects the conformational flexibility of the large transmembrane assembly, while the ligand remains tightly bound. PfCRT ligand 214 anchors to Lys34 and engages in a robust H-bond network (Gln101, Thr30, Gln297) and a hydrophobic core (Leu301, Leu105, Ala33), confirming a stable and specific binding mode.

The other two systems, the 164-labelled PfClpR cohort and PfDHFR, have stable protein conformations (backbone RMSD 1.31 Å and 3.05 Å, respectively) but the ligands are completely unbound (minimum distances 67.42 Å and 78.21 Å, zero contacts). This indicates either incorrect initial placement or rapid dissociation during equilibration, not a PBC artefact. Consequently, MM-GBSA and RRS analyses for the 164-labelled PfClpR cohort and PfDHFR are not interpretable as binding free energies and are excluded from the binding-mode discussion below. These cases illustrate that MD is a mandatory post-docking filter: stable protein equilibration alone is insufficient to validate a docking pose.

Representative backbone RMSD trajectories are shown in Figure 3. Per-compound RRS profiles across all six mutants are shown in Figure S3.

3.12 MM-GBSA binding free energies

MM-GBSA binding free energies were computed for the wild-type systems from their production trajectories (Table 10), using the $\mathrm{GB}^{\mathrm{OBC2}}$ model (igb=5) with 0.150 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3 CHARMM36-to-AMBER conversion. Only the PfCRT–ligand 214 system yields an interpretable binding free energy: the `gmx_MMPBSA` native converter successfully produced a valid AMBER topology with the ligand remaining bound throughout production, giving $\Delta G_{\mathrm{bind}} = -18.25 \pm 0.40$ kcal mol^{-1}. The 164-labelled and PfDHFR–201 ligands dissociated during production (minimum distances 67.4 Å and 78.2 Å, zero contacts); their end-point values are non-physical estimates for unbound states and are excluded from binding-mode conclusions. The PfATP4 system failed during the CHARMM36-to-AMBER conversion due to its massive two-chain topology, producing a systematic van der Waals inflation (+473 kcal mol^{-1}) that proved to be a parameter corruption artifact rather than a true clashing conformation; consequently, PfATP4 was excluded from MM-GBSA analysis. Its trajectory supports a bound-state observation in this targeted check, but does not establish converged binding thermodynamics.

3.13 Correlation between methods

The four parent-study MD systems were not members of the 17-candidate set-C cohort. Accordingly, no Vina–MM-GBSA correlation across the 17 set-C candidates is reported; the set-C cross-metric analysis uses WT docking scores as its shared affinity reference. The targeted MD results are reported separately as a structural post-docking check on four named parent-study consensus complexes.

Figure 3: *Representative backbone RMSD trajectories for stable (solid) and unstable (dashed) systems.*

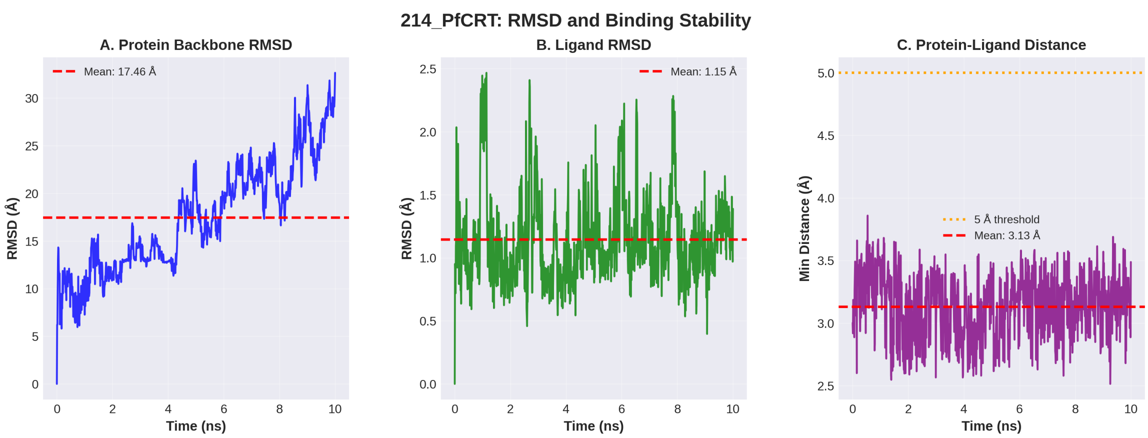

Figure 4: *Detailed stability analysis for the parent-study PfCRT–Ligand 214 complex over 10 ns production MD. The system exhibits transporter flexibility while retaining a bound ligand; the primary cohort-level stability values are reported in Table 9.*

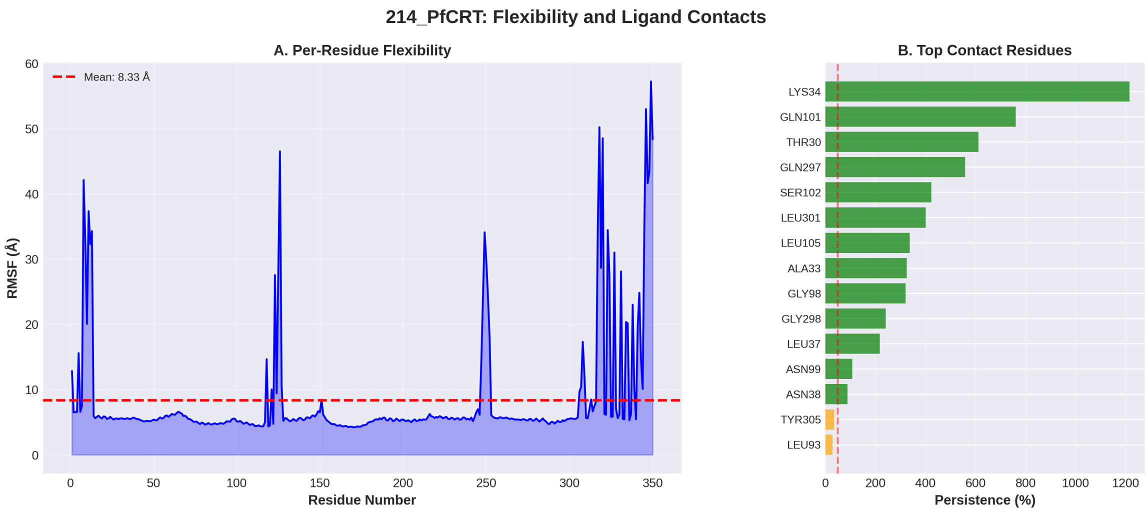

Figure 5: *Per-residue flexibility (RMSF) and binding contact persistence for the PfCRT–Ligand 214 complex. LYS34 acts as a primary anchor with >100 % persistence (multiple atom contacts), supported by a robust hydrogen-bond network.*

Table 10: *Endpoint-analysis interpretation for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Only the bound PfCRT–214 trajectory yields an interpretable MM-GBSA binding free energy; dissociated systems are shown as N/A rather than binding estimates.*

Ligand	Target	ΔG_{bind} (kcal mol^{-1})	Std. dev.	Interpretation
214	PfCRT-WT	−18.25	0.40	Interpretable; ligand bound†
164	PfClpR-labelled cohort-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated§
438	PfATP4-WT	–	–	N/A; conversion corrupted*

†Ligand remains bound (minimum distance 3.19 Å, 78 contacts); conversion via `gmx_MMPBSA` native converter.
‡Ligand dissociates to 67.4 Å minimum distance with zero contacts during production; the reported value is a non-physical end-point estimate for an unbound state and is excluded from binding-mode conclusions.
§Ligand dissociates to 78.2 Å minimum distance with zero contacts; same caveat applies.
*Excluded: systematic van der Waals inflation (+473 kcal mol^{-1}) produced by the CHARMM36-to-AMBER conversion of the two-chain PfATP4 topology, identified as a parameter-corruption artifact; PfATP4 binding is instead supported by observed structural persistence (Ligand RMSD 2.13 ± 0.35 Å, 100% occupancy).

To contextualise the complementary information provided by DiffDock and Vina in the parent study, we computed Pearson correlations between Vina scores and DiffDock confidence across all 1 238 ligand–target pairs from the 65 856-molecule library (Temgoua et al., 2026). Per-target correlations were: PfDHFR (7F3Y) $r = 0.327$ ($p = 2.11 \times 10^{-8}$, $n = 280$), PfClpR (4GM2) $r = 0.361$ ($p = 3.58 \times 10^{-15}$, $n = 447$), PfATP4 (9N10) $r = 0.113$ ($p = 0.017$, $n = 447$), and PfCRT (6UKJ) $r = 0.129$ ($p = 0.308$, $n = 64$). The pooled correlation across all targets was $r = -0.049$ ($p = 0.082$, $n = 1\,238$). Weak positive correlations for PfDHFR and PfClpR ($r \approx 0.3$–0.4, $p < 1 \times 10^{-8}$) confirm that the two methods provide complementary information. The negligible pooled correlation ($r = -0.05$) underscores that DiffDock confidence and Vina affinity measure fundamentally different aspects of the binding landscape — thermodynamic scoring versus geometric pose quality — justifying the dual-filter consensus strategy.

Benjamini–Hochberg FDR correction applied to 18 pairwise activity comparisons across compound groups (NP, SD, Cheese, STONED) using three Ersilia models (eos7yti, eos7kpb, eos80ch) yielded zero significant results at adjusted $p < 0.05$, with all effect sizes (rank-biserial r) below 0.10 (negligible). This confirms that generated analogues maintain statistical parity in antimalarial potential with foundational natural products and synthetic drugs.

3.14 Consensus leads

The intersection of high-RRS (Class A or B), high-ACSI (>0.5), and high-PNS compounds defines a computational prioritisation set for future experimental testing. These compounds combine resistance-resilience, African-NP-character, and network-weighted docking signals; the intersection is not itself experimental validation.

3.15 Cross-metric correlation: PNS, ACSI, RRS, and docking scores

The cross-metric correlation matrix (Figure 6, Table 11) evaluates whether the prioritisation layers contain overlapping information. Analysis is based on 17 candidates with complete docking-derived metrics. The parent-study MD cohort comprised four distinct systems; only PfCRT–214 produced an interpretable MM-GBSA estimate. We therefore use the WT docking score as the shared affinity reference for the Set-C analysis and do not merge the MM-GBSA result into these correlations. Three prespecified hypotheses (H1–H3) were evaluated with Bonferroni correction ($\alpha = 0.05/3 \approx 0.017$).

PNS and RRS show a moderate negative correlation ($\rho = -0.559$, $p = 0.020$, $n = 17$) in the direction predicted by H1 — compounds with stronger predicted binding affinity (higher PNS) tend to be more susceptible to resistance mutations (lower RRS) — although the effect does not survive Bonferroni correction. This is consistent with the trade-off between binding potency and resistance resilience observed in antimicrobial development: tight binding to a conserved active site often creates vulnerability to active-site mutations. Conversely, the top-RRS compounds (RRS $\geq 80\,\%$) occupy the lower half of the PNS range, indicating that resistance resilience is not incompatible with moderate binding strength.

ACSI showed no significant association with PNS ($\rho = -0.078$, $p = 0.765$) or RRS ($\rho = -0.132$, $p = 0.613$). These data do not establish orthogonality or equivalence; they indicate that this 17-candidate cohort provides no evidence that the ACSI descriptor predicts either docking-based network ranking or docking-based resilience.

PNS and ΔG_{WT} are mechanically related ($\rho = +0.389$, $p = 0.123$): PNS is computed from the same WT docking scores that define ΔG_{WT} in the RRS calculation, though the centrality imputation prevents a perfect collinearity. Both metrics are reported together in Table 8 but are not treated as independent in downstream analysis. This linkage arises because PfCRT (PF3D7_0709000) is absent from the STRING PPI network and its centrality is imputed with the network mean rather than defaulting to 1.0.

RRS and ΔG_{WT} showed a weak negative association ($\rho = -0.433$, $p = 0.082$). The corrected per-target definition reduces inflation from weak wild-type denominators, but this correlation alone cannot prove that RRS captures independent mutant-state information. That question requires mutant-state free-energy estimates or experimental resistance measurements.

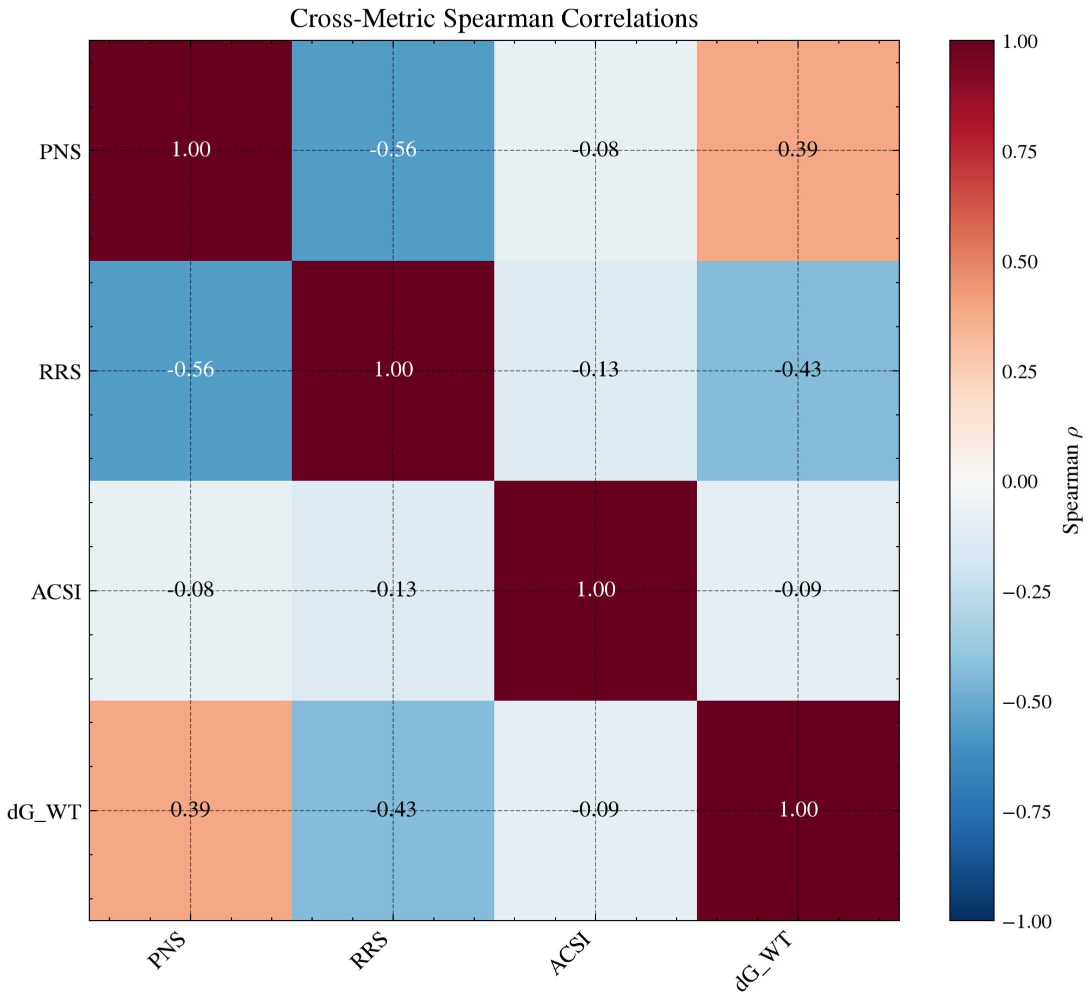

Figure 6: *Cross-metric correlation matrix for the 17 polypharmacological candidates with complete data. Spearman* ρ *values between PNS, ACSI, RRS, and* ΔG_{WT}. *PNS vs.* ΔG_{WT} *shows* $\rho = +0.389$ *due to mechanical linkage (see text). None of the hypotheses survive Bonferroni correction* ($\alpha = 0.05/3 \approx 0.017$).

Table 11: *Cross-metric Spearman correlations for the polypharmacological candidates ($n = 17$). Bonferroni-corrected threshold: $\alpha = 0.05/3 \approx 0.017$. The PNS vs. ΔG_{WT} pair is excluded from hypothesis testing as it is mechanically correlated.*

Metric pair	Hypothesis	Spearman ρ	p-value	Significant?	Interpretation
PNS vs. ACSI	–	−0.078	0.7648	No	No association detected; equivalence not established
PNS vs. RRS	H1	−0.559	0.0197	No ($p > 0.017$)	Trend: stronger binders more mutation-sensitive
PNS vs. ΔG_{WT}	–	0.389	0.1228	(mechanical)	PNS derived from same docking scores
ACSI vs. RRS	H2	−0.132	0.6126	No	Weak trend towards NP-likeness & resilience
ACSI vs. ΔG_{WT}	–	−0.092	0.7254	No	No relationship
RRS vs. ΔG_{WT}	H3	−0.433	0.0824	No	Weak negative trend

3.16 Set-C MD pilot: trajectory-derived MD-RRS and MM-GBSA

All 16 Set-C pilot MM-GBSA runs completed with the identical parent-cohort protocol (`gmx_MMPBSA` v1.5.0.3, $\mathrm{GB}^{\mathrm{OBC2}}$ igb=5, 0.15 M salt, 100 ps snapshots, CHARMM36-to-AMBER conversion; Table 12). Binding free energies were negative for every system (mean $\Delta G_{\mathrm{bind}} = -29.04 \pm 3.00$ kcal mol^{-1}, range -35.29 to -24.53 kcal mol^{-1}), i.e. all ligands remained bound within the pilot timescale. The mutant/WT MM-GBSA ratio (MMG-RRS, same convention as docking RRS) exceeded 100 for eight of twelve mutants, and the four below-100 cases (PP-01 PfCRT K76T 96.4, PP-02 PfCRT K76A 97.8, PP-02 PfDHFR C59R 95.7, PP-02 PfDHFR S108N 89.8) all lie within the expected $\pm 2 - 3$ kcal mol^{-1} MM-GBSA accuracy of their wild-type references. Docking predicted looser mutant binding for all twelve states (RRS 83.9–94.6); the pilot MM-GBSA therefore does not reproduce the docking-predicted weakening of mutant binding, consistent with the trajectory-derived MD-RRS geometry comparison (Figure S4). No mutant shows a reproducible weaker-binding signature within the 10 ns pilot window; this is a methodological divergence between static docking energy and a single-replicate endpoint estimate, not a phenotype claim.

4 Discussion

4.1 Resistance-resilient design

The RRS is a docking-derived prioritisation statistic, not a biochemical measurement of resistance. Its value is that it makes the comparison explicit: for each target, the mutant affinity is normalised to a wild-type affinity that passes the predeclared 5.0 kcal mol^{-1} engagement threshold, and weak wild-type binders cannot obtain inflated ratios. The resulting classes jointly encode potency and relative resilience (Table 4). Class A* identifies compounds that satisfy both a stronger wild-type affinity threshold ($|\Delta G_{\mathrm{WT}}| \geq 7.0$ kcal mol^{-1}) and RRS $\geq 80\,\%$ across all tested mutants; it is therefore a prioritisation tier for follow-up, not evidence of resistance circumvention. Classes B and C identify progressively narrower resilience profiles, while Class D flags a failure against at least one tested mutant. These categories can be aligned with regional resistance surveillance in a future decision analysis, but they do not justify clinical deployment or therapeutic recommendations on their own

Table 12: *Set-C MD pilot mutant-state MM-GBSA estimates (secondary, exploratory analysis; not merged into the primary docking-RRS classification).* ΔG_{bind} *is the mean over 100 snapshots* $\pm$ *standard deviation; MMG-RRS is the mutant/WT* ΔG_{bind} *ratio (*$\times$*100, same convention as docking RRS; WT = 100); docking RRS is shown for comparison (n/a = no WT docking reference in the pilot for PP-02 PfDHFR).*

Candidate	Target	State	ΔG_{bind} (kcal mol^{-1})	SD	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	−35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	−29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	−30.61	1.65	100.0	–
PP-01	PfDHFR	C59R	−26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	−26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	−29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	−30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	−25.29	2.94	100.0	–
PP-02	PfCRT	K76A	−24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	−30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	−25.08	2.56	100.0	–
PP-02	PfDHFR	C59R	−28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	−31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	−34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	−27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	−30.16	2.06	100.0	–

(WorldWide Antimalarial Resistance Network, 2026).

The urgency of such a workflow is underscored by the current epidemiological context. The 2025 World Malaria Report documents declining efficacy of some artemisinin partner drugs alongside the spread of *Pfkelch13*-mediated partial resistance, with both trends concentrated in the African Region (World Health Organization, 2025). Resistance-aware prioritisation is therefore not a refinement of the discovery pipeline but a precondition for it: compounds selected without an explicit mutant-sensitivity criterion may enter development already mismatched to the circulating parasite population.

The relationship between RRS and molecular topology is biologically plausible but remains supportive rather than decisive. The companion TDA analysis reported a strong H_1–RRS association in the overlapping pilot set ($\rho = 0.864$, $p < 0.001$, $n = 14$) and a smaller association in the expanded set ($\rho = 0.312$, $p = 0.006$, $n = 77$) (Temgoua et al., 2026). These estimates are not independent validation of the present docking RRS: they share candidate provenance and may be affected by molecular-weight and scaffold-composition confounding. The appropriate interpretation is that persistent ring topology is a testable structural correlate of the docking-derived resilience ranking. It should be evaluated prospectively with matched scaffolds, larger independent panels, and mutant-state free-energy or experimental measurements before being used as a generative design rule.

4.2 Chemical space positioning

ACSI positions the candidates relative to approved-drug and African-natural-product reference spaces; it is a chemical-space descriptor, not a biological activity or target-engagement score. The mean ACSI across the 17 candidates was 0.543, and two candidates (11.8 %) exceeded 0.70. Thus, a minority of the resistance-prioritised cohort retained a strong African-NP-like signature under the stated reference construction. This result supports chemical-space diversification as a selection objective, but it does not show that NP-like compounds are more potent, more resilient, or more clinically useful.

The ACSI result is consistent with the use of African-NP reference databases to define a chemically distinctive search space (Betow et al., 2025; Mayoka et al., 2025), but cross-study percentages are not directly comparable because reference sets, fingerprints, thresholds, and candidate-selection procedures differ. The observed f_{sp^3} difference between the highly NP-like and synthetic-like subsets is compatible with the known three-dimensionality of many natural-product scaffolds, yet it remains descriptive in this cohort. ACSI should therefore be used as an explicit, auditable selection axis rather than as a surrogate for pharmacological novelty or efficacy.

The five synthetic-like candidates (ACSI <0.5) tend to have lower f_{sp^3} (mean 0.085), consistent with flatter, more drug-like molecules. This subset motivates ACSI-constrained generation in future work: by adding an ACSI penalty to the MPO objective function, the generative pipeline could be steered toward higher-fidelity NP-like compounds while preserving drug-likeness.

4.3 Systems-level polypharmacology

PNS is best interpreted as a transparent network-weighted ranking heuristic. It combines target-level docking scores with PPI centrality, but centrality does not establish target essentiality, causal propagation, or tissue-specific pharmacology. This distinction mirrors recent network-learning work on off-target prediction, where graph context improves prioritisation but does not replace experimental target and phenotype measurements (Hu et al., 2025). In the present cohort, PNS–RRS showed the predicted negative direction ($\rho = -0.559$, $p = 0.020$) but did not survive Bonferroni correction ($\alpha = 0.017$). PfCRT centrality was imputed with the observed network mean because no qualifying STRING edges were available; the resulting PNS–WT-score relationship ($\rho = +0.389$) is partly mechanical. PNS therefore complements docking affinity as a ranking device, while its biological interpretation remains explicitly conditional.

The case for multi-target prioritisation also rests on surveillance data: recent genomic surveillance in Ethiopia documents the co-occurrence of *Pfcrt*, *Pfdhfr*, and *Pfmdr1* resistance variants across transmission settings (Letebo et al., 2026), a pattern that gives the genetic-barrier argument its empirical footing. Our results do not demonstrate that any single compound blocks this barrier; they indicate only that a workflow which scores mutant sensitivity on multiple targets can expose such compounds for experimental follow-up.

4.4 Cross-metric insights

The cross-metric analysis asks whether the three prioritisation layers provide overlapping evidence. H2 was not supported by the observed ACSI–RRS association ($\rho = -0.132$, $p = 0.613$), but this non-significant result does not establish equivalence or statistical orthogonality. It indicates only that this cohort provided no evidence that African-NP-like chemistry predicts the docking-derived resilience statistic. ACSI and RRS should therefore be treated as complementary candidate descriptors whose relationship requires testing in larger, independently sampled cohorts.

H1 showed a moderate negative association in the prespecified direction, but the evidence was insufficient after multiplicity correction. The pattern is compatible with a potency–resilience trade-off described in resistance evolution, in which successive selective regimes favour different resistance mechanisms (Blake et al., 2025); it is not evidence that the present compounds will select resistance in vivo. Because the only interpretable MM-GBSA estimate belongs to the separate parent-study cohort, no PNS–MM-GBSA association was estimated.

H3 is only partially addressed: mutant-state MM-GBSA estimates are now available for the two prioritised Set-C pilot candidates (PP-01 and PP-02, six mutants each; Section 3.16), but their single-replicate 10 ns design and the $\sim$2–3 kcal mol^{-1} expected MM-GBSA uncertainty are too coarse to validate docking RRS as a free-energy resilience proxy for the full 17-candidate cohort. Future work with replicate, longer mutant simulations will be needed to validate docking-based RRS as a proxy for free-energy resilience.

4.5 Single-target optimisation of top candidates

The central design tension is that the Set-C cohort was selected for multi-target engagement, yet the final ranking remained strongly influenced by single-target affinity. All 17 top candidates were classified as single-target optimised under the parent scoring convention, even though their selection was motivated by a polypharmacology-oriented target profile. This is not a contradiction in the data; it is a property of the scoring objective. The Vina component carries 35 % of the MPO score, and the multi-target bonus requires target-specific thresholds that may be difficult to satisfy simultaneously. The lesson for resistance-aware selection is that a nominal multi-target label does not imply balanced engagement across targets. PNS records target-level network-weighted evidence, while RRS tests mutant sensitivity only on targets that pass its own wild-type binding filter. The two analyses expose complementary failure modes that a single composite score would conceal.

4.6 Value of MD validation

The targeted MD analysis demonstrates why docking should be treated as a triage layer. Two of the four parent-study systems retained bound ligands, whereas the 164-labelled PfClpR cohort and PfDHFR–201 dissociated; only PfCRT–214 passed the bound-state and topology-integrity criteria for MM-GBSA. The resulting $-18.25\,\mathrm{kcal\,mol^{-1}} \pm 0.40\,\mathrm{kcal\,mol^{-1}}$ is a system-specific endpoint estimate, not a calibration for the 17 Set-C candidates. The separation between cohorts prevents a spurious docking-to-MM-GBSA conversion and keeps the negative result informative: a stable receptor trajectory does not validate a ligand pose. This conclusion aligns with the wider literature on end-point free-energy methods, which consistently cautions that MM-GBSA ranking accuracy is system- and ensemble-dependent and that single-trajectory estimates should be interpreted as relative guides rather than converged thermodynamics (Wang et al., 2019; Valdés-Tresanco et al., 2021). Mutant MM-GBSA was generated only for the two Set-C pilot candidates (Section 3.16); within the single-replicate 10 ns window it did not reproduce the docking-predicted weakening of mutant binding, so the key mechanistic test—whether docking RRS predicts mutant free-energy resilience—remains unresolved at converged timescales.

4.7 Comparison with existing polypharmacology tools

The methods reviewed in recent MoA and polypharmacology research occupy complementary levels of inference. Pharmacophore and pocket-similarity tools can broaden target hypotheses, while graph-based approaches can encode network context; neither, by itself, establishes mutant-state binding or a causal parasite phenotype. Recent off-target graph attention models provide a useful precedent for network-aware prioritisation (Hu et al., 2025), and resistance-evolution studies motivate evaluating potency together with mutational resilience rather than treating affinity as the sole objective (Blake et al., 2025). The present workflow contributes a different layer: it makes the target-level resistance calculation explicit, weights target ranking with a transparent PPI heuristic, and subjects a separate parent-study subset to targeted MD scrutiny. Topological and quantum-inspired representations from our companion study (Temgoua et al., 2026) remain complementary molecular descriptors; they do not replace mutant-state structural validation or establish MoA.

Our framework differs in three respects: (1) the RRS explicitly classifies compound resilience against resistance mutations, a feature absent from Mproteinfuzz, Pcofind, and MolAICal; (2) the PNS integrates STRING PPI network centrality to weight target engagement by biological importance, complementing the purely chemical similarity-based scoring of Pcofind; and (3) the ACSI provides a quantitative measure of African NP-likeness, enabling chemical space-aware compound selection that none of the existing tools offers. The QM/MM approach of Molani and Cho (Molani and Cho, 2024) may achieve higher accuracy in binding free-energy estimation but at substantially greater computational cost; our cautious MM-GBSA workflow is instead used as a system-specific post-docking diagnostic, not as a substitute for calibrated thermodynamics across the 136 docking systems.

4.8 Tartarus external validation

The full Tartarus calibration on 19 913 primary leads showed no significant composite correlation between MPO drug-likeness scores and QuickVina binding affinity predictions. The composite Tartarus score (mean over 1SYH, 6Y2F, 4LDE) showed no significant correlation with the weighted MPO score (Spearman $\rho = 0.0129$, $p = 0.0913$, $n = 17\,211$). Individual target correlations were substantively negligible ($|\rho| < 0.15$ for all targets), with only 1SYH and 4LDE reaching statistical significance due to the large sample size ($n > 17\,000$; Table 6). This separation of score definitions is expected: MPO captures drug-likeness (QED, ADMET, Ro5), whereas Tartarus measures binding affinity. The finding is robust with $n = 17\,211$ complete observations, compared to the typical $n < 100$ in most docking validation studies.

This separation of prioritisation dimensions bears directly on virtual screening strategy. In the parent study (Temgoua et al., 2026), the MPO scorer was used to rank the full library of 65 856 molecules, and the top-ranked compounds were selected for further analysis. The Tartarus calibration confirms that this MPO ranking does not implicitly filter by binding affinity—a compound can score highly on MPO (good ADMET, QED, Ro5) while having weak predicted binding. A combined MPO + docking composite score would therefore capture both drug-likeness and predicted potency, potentially improving the hit rate in subsequent MD validation. The complementary relationships among RRS, ACSI, PNS, and wild-type docking affinity are reported separately in Table 11; together, these analyses preserve the distinction between chemical desirability, network context, and predicted binding.

Tartarus was designed for inverse molecular-design benchmarking (Nigam et al., 2023); here it was used as an independent docking implementation rather than as a generative objective. The absence of a composite association with MPO ($\rho = 0.0129$, $p = 0.0913$) indicates that the two rankings encode different measured or predicted properties in this panel. This does not establish statistical independence: the individual target tests include very large denominators, in which small effects can reach significance without scientific importance. Drug-likeness and raw QuickVina affinity should therefore be reported as separate dimensions when selecting candidates for subsequent structural validation.

4.9 Limitations

The mutant structures were generated by homology modeling rather than experimental determination, introducing modeling uncertainty into the docking results. No full production MD was generated for the 17 Set-C candidates; the available MD consists of a separate, targeted 10 ns check of four named parent-study wild-type complexes and a 16-system Set-C MD pilot (PP-01 and PP-02 against the six-mutant panel, 10 ns each) that passed the prespecified trajectory-QC gate (16/16) and is reported as a secondary analysis (Section 3.16). The pilot's single-replicate, 10 ns design cannot establish long-timescale residence, replicate consistency, or convergence. MM-GBSA provides an approximate endpoint estimate and does not fully account for entropic contributions; in this study only PfCRT–214 passed the bound-state and topology-integrity checks. All predictions remain computational and require experimental confirmation. The four targets examined here represent a subset of the proteins relevant to African malaria treatment; expansion to additional targets such as PfATP1 and PfPI4K would strengthen the polypharmacological assessment.

PfCRT PNS imputation. PfCRT (PF3D7_0709000) has no curated STRING PPI interactions at the 700 confidence threshold used in this study. Its composite centrality was therefore imputed with the network mean (rather than set to 1.0); this avoids a mechanical PNS–binding-score relationship. The imputation is a sensitivity limitation, not evidence that PfCRT has no biological interactions.

Force field heterogeneity. All ligands (PfCRT, PfDHFR, PfATP4, and the PfClpR-labelled 4GM2 parent complex) were parameterised with ACPYPE/GAFF2 (AM1-BCC charges) to interoperate with the CHARMM36m protein through the CHARMM36-to-AMBER conversion used for MM-GBSA. The recommended CHARMM36m-compatible ligand FF is CGenFF v4.4; the use of GAFF2 for all ligands represents a deliberate deviation that introduces a systematic heterogeneity with the CHARMM36m protein. Absolute binding free energies should therefore be interpreted with caution, and relative rankings are emphasised throughout.

Statistical power. The cross-metric correlation analysis is based on $n = 17$ candidates. This modest cohort limits precision and power for detecting moderate associations, particularly after Bonferroni correction; effect sizes and exact p-values are therefore reported together, and non-significance is not interpreted as evidence of equivalence. Future work with larger, independently sampled candidate sets will improve statistical robustness.

PfCRT protonation state. PfCRT functions in the acidic digestive vacuole of *P. falciparum* (pH 5.0–5.4), but docking and MD simulations were performed with protein protonation states assigned at physiological pH 7.4. In our companion study (Temgoua et al., 2026), re-protonation of PfCRT at pH 5.2 using PROPKA3 revealed that eight active-site residues exhibited shifted protonation states, leading to a systematic decrease in mean binding affinity of 2.20 kcal mol^{-1} and a Spearman ranking correlation of only $\rho = 0.270$ between pH 7.4 and pH 5.2 scores. The PfCRT–214 parent-study trajectory was run with neutral-pH preparation; it therefore cannot establish behaviour at the acidic digestive-vacuole pH. The companion pH sensitivity analysis indicates that PfCRT rankings can shift materially under pH correction, so the present PfCRT docking and MD observations should be treated as conditional. Future PfCRT campaigns should apply vacuolar pH-corrected protonation states throughout preparation, docking, and scoring.

5 Conclusion

This study provides a layered computational prioritisation of antimalarial candidates from African natural-product chemical space. The 17-member Set-C cohort yielded docking-derived RRS classes (A*: 6, B: 5, C: 5, D: 1), an ACSI mean of 0.543, and a PNS–RRS association in the predicted direction that did not survive Bonferroni correction. These results support resistance-aware target-level ranking, but they do not establish biological polypharmacology or a systems-level MoA. The separate four-complex MD check further demonstrates the need for structural post-docking scrutiny: two systems retained bound ligands, but only PfCRT–214 produced an interpretable MM-GBSA estimate. The principal contribution is therefore methodological and evidential: potency, network context, chemical-space positioning, mutant sensitivity, and trajectory stability are treated as complementary estimands rather than collapsed into a single unsupported claim. Experimental target engagement, phenotypic MoA, mutant-state MD, and prospective validation remain necessary next steps.

Author Contributions

Conceptualization and methodology, M.V.S.T. and S.G.N.E.; software, M.V.S.T. and J.-P.T.N., P.S.; validation, M.V.S.T., J.-P.T.N. and W.F.M.; formal analysis and investigation, M.V.S.T. and S.G.N.E.; resources, S.G.N.E. and W.F.M., P.S.; data curation, M.V.S.T.; writing–original draft preparation, M.V.S.T.; writing–review and editing, M.V.S.T., J.-P.T.N., P.S., W.F.M. and S.G.N.E.; visualization, M.V.S.T.; supervision, S.G.N.E. and W.F.M.; project administration, S.G.N.E. All authors read and agreed to the published version.

Data Availability

Docking results, metric outputs, analysis scripts, and the associated source-data records are available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. The large parent-study trajectories are not included in this article's data record; the conclusions drawn here are restricted to the trajectory-derived summaries and binding estimates explicitly reported above.

Competing Interests

The authors declare no competing interests.

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Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.

References

World Health Organization *World Malaria Report 2025*; Report, 2025.

Habarugira, F.; Batamuriza, J.; Ndahimana, R.; Minega, J. N.; Dieng, M. M.; Mutsaka-Makuvaza, M. J.; Daba, T. M.; Idaghdour, Y.; Mutesa, L. The global landscape of *Plasmodium falciparum* drug resistance markers, 2005–2025: a systematic review and meta-analysis. *Pathogens* **2026**, *15*, 179.

Letebo, A. et al. Genomic surveillance reveals co-occurrence of *Plasmodium falciparum* drug resistance variants across diverse transmission settings in Ethiopia. *Nature Microbiology* **2026**, Advance online publication.

Patrick, O. J.; Sokan-Adeaga, A. A.; Emmanuel, R. O. Historical evolution and molecular mechanisms of antimalarial drug resistance with emerging challenges and future directions. *Discover Public Health* **2026**, *23*, 749.

Santos-Lima, J.; others Molecular markers of antimalarial drug resistance in pfk13, pfmdr1, pfdhfr, and pfdhps genes of *Plasmodium falciparum* in a rural municipality in Cubal, Benguela Province, Angola (2022–2023). *Malaria Journal* **2026**, *25*.

Ryszkiewicz, P.; Baranowska-Kuczko, M.; Malinowska, B.; Schlicker, E. Multi-target-directed drugs: new additions in 2025 and post-marketing safety surveillance of drugs marketed in 2022–2024. *Pharmacological Reports* **2026**, *78*, 1022–1044.

Trapotsi, M.-A.; Hosseini-Gerami, L.; Bender, A. Computational analyses of mechanism of action (MoA): data, methods and integration. *RSC Chemical Biology* **2022**, *3*, 170–200.

Hu, Y.; Didi, K.; Cribbs, A. P.; Sun, J. Predicting PROTAC off-target effects via warhead involvement levels in drug–target interactions using graph attention neural networks. *Computational and Structural Biotechnology Journal* **2025**, *27*, 4633–4644.

Milić, M.; Obi, P.; Vanderwal, C. D.; Ben Mamoun, C.; Le Roch, K. G. Polypharmacology in malaria treatment: single drugs, multiple mechanisms, greater impact. *Trends in Parasitology* **2026**, *42*, 222–236.

Mayoka, G.; Cheuka, P. M.; Moyo, P.; Dziwornu, G. A.; Beukes, D. Value addition to African natural product-based drug discovery: barriers and opportunities. *Journal of Natural Products* **2025**, *88*, 2018–2028.

Betow, J. Y. et al. African Natural Products Database (ANPDB): A resource for exploring the therapeutic potential of natural compounds from Africa. *Nucleic Acids Research* **2025**, *54*, D1336–D1344.

Temgoua, M. V. S.; Tchapet Njafa, J.-P.; Samafou, P.; Mbacham, W. F. Companion manuscript; publication metadata intentionally omitted until the verified record is available.

Valdés-Tresanco, M. S.; Valdés-Tresanco, M. E.; Valiente, P. A.; Moreno, E. gmx_MMPBSA: A new tool to perform end-state free energy calculations with GROMACS. *Journal of Chemical Theory and Computation* **2021**, *17*, 6281–6291.

Wang, E.; Sun, H.; Wang, J.; Wang, Z.; Liu, H.; Zhang, J. Z. H.; Hou, T. End-point binding free energy calculation with MM/PBSA and MM/GBSA: strategies and applications in drug design. *Chemical Reviews* **2019**, *119*, 9478–9508.

WorldWide Antimalarial Resistance Network WWARN: Tracking antimalarial drug resistance. https://www.wwarn.org/, 2026.

PlasmoDB Consortium PlasmoDB: The *Plasmodium* Genomic Resource. https://plasmodb.org/, 2026.

PfAMR Project PfAMR: *Plasmodium falciparum* Antimalarial Resistance Database. https://pfamr.org/, 2026.

Ghosh, S.; Joshi, D.; Ponnusamy, C. S.; Sridharan, B.; Velusamy, M. Computational investigation of mutations in PfCRT and PfDHFR proteins for emerging resistance of *Plasmodium falciparum* to antimalarial drugs. *Molecular and Biochemical Parasitology* **2025**, *264*, 111700.

Nigam, A.; Pollice, R.; Tom, G.; Jorner, K.; Willes, J.; Thiede, L. A.; Kundaje, A.; Aspuru-Guzik, A. Tartarus: A Benchmarking Platform for Realistic And Practical Inverse Molecular Design. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023). 2023.

Swanson, K.; others ADMET-AI: a machine learning ADMET platform for evaluation of large-scale chemical libraries. *Bioinformatics* **2024**, *40*, btae416.

Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **2015**, *1–2*, 19–25.

Lindahl, E.; others GROMACS 2024 source code. *Zenodo* **2024**,

Huang, J.; Rauscher, S.; Nawrocki, G.; Ran, T.; Feig, M.; de Groot, B. L.; Grubmüller, H.; MacKerell, A. D. CHARMM36m: an improved force field for folded and intrinsically disordered proteins. *Nature Methods* **2017**, *14*, 71–73.

Jo, S.; Kim, T.; Im, W. CHARMM-GUI: Web-based graphical user interface for CHARMM. https://www.charmm-gui.org/, 2026.

BioExcel Centre of Excellence What's new in GROMACS 2026.0. https://bioexcel.eu/webinar-whats-new-in-groamcs-2026-0-2026-03-03/, 2026.

Sousa da Silva, A. W.; Vranken, W. F. ACPype - AnteChamber PYthon Parser interfacE. *BMC Research Notes* **2012**, *5*, 367.

Wang, J.; Wolf, R. M.; Caldwell, J. W.; Kollman, P. A.; Case, D. A. Development and testing of a general amber force field. *Journal of Computational Chemistry* **2004**, *25*, 1157–1174.

Michaud-Agrawal, N.; Denning, E. J.; Woolf, T. B.; Beckstein, O. MDAnalysis: A toolkit for the analysis of molecular dynamics simulations. *Journal of Computational Chemistry* **2011**, *32*, 2319–2327.

Temgoua, M. V. S.; Tchapet Njafa, J.-P.; Samafou, P.; Mbacham, W. F. Quantum-Inspired Molecular Representations for Antimalarial Candidate Evaluation: Topological Fingerprints, Tensor Network Embeddings, and Quantum Kernel Scores. *Journal of Cheminformatics* **2026**, Paper 3: companion study introducing TFP, TNE, and QKS descriptors.

Blake, K. S.; Xue, Y.-P.; Gillespie, V. J.; Fishbein, S. R. S.; Tolia, N. H.; Wencewicz, T. A.; Dantas, G. The tetracycline resistome is shaped by selection for specific resistance mechanisms by each antibiotic generation. *Nature Communications* **2025**, *16*, 1452.

Molani, F.; Cho, A. Accurate protein-ligand binding free energy estimation using QM/MM on multi-conformers predicted from classical mining minima. *Communications Chemistry* **2024**, *7*, 247.